ARCHIVELOG

- **How to set recovery file destination to '/u01/app/oracle/FRA'**

STEP 1: Check the recovery file destination.

```
SQL>
SQL> SELECT NAME,space_limit/1024/1024 AS "SIZE_MB",space_used/1024/1024 AS "USED_MB",space_reclaimable/1024/1024 AS "RECLAIM_MB" FROM V$RECOV
ERY_FILE_DEST;

NAME                                                                SIZE_MB    USED_MB RECLAIM_MB
------------------------------------------------------------------ ---------- ---------- ----------
/u01/app/oracle/fast_recovery_area                                       8256 4154.22705 1052.83594
```

STEP 2: Check FRA directory is present or not in OS level location

```
[oracle@localhost oracle]$ pwd
/u01/app/oracle
[oracle@localhost oracle]$ ls
admin  audit  c1.sql  cfgtoollogs  checkpoints  diag  fast_recovery_area  oradata  product
[oracle@localhost oracle]$
```

STEP 3: Create a directory in OS level

```
[oracle@localhost oracle]$ mkdir FRA
[oracle@localhost oracle]$
[oracle@localhost oracle]$ ls
admin  audit  c1.sql  cfgtoollogs  checkpoints  diag  fast_recovery_area  FRA  oradata  product
[oracle@localhost oracle]$
```

STEP 4: Set FRA location

```
SQL> ALTER SYSTEM SET db_recovery_file_dest='/u01/app/oracle/FRA' SCOPE=BOTH;

System altered.

SQL> SELECT NAME,space_limit/1024/1024 AS "SIZE_MB",space_used/1024/1024 AS "USED_MB",space_reclaimable/1024/1024 AS "RECLAIM_MB" FROM V$RECOV
ERY_FILE_DEST;

NAME                                                                SIZE_MB    USED_MB RECLAIM_MB
------------------------------------------------------------------ ---------- ---------- ----------
/u01/app/oracle/FRA                                                      8256 4154.22705 1052.83594
```

- **How to set recovery file destination Size to 8900MB.**

STEP 1: Check the file destination size

```
SQL> SELECT NAME, SPACE_LIMIT/1024/1024 AS "FILE_DESTINATION_SIZE_MB" FROM V$RECOVERY_FILE_DEST;

NAME                                                              FILE_DESTINATION_SIZE_MB
----------------------------------------------------------------- ------------------------
/u01/app/oracle/fast_recovery_area                                                    8256
```

STEP 2: Change the size of the file destination

```
SQL> ALTER SYSTEM SET db_recovery_file_dest_size=8900M SCOPE=BOTH;

System altered.

SQL> SELECT NAME, SPACE_LIMIT/1024/1024 AS "FILE_DESTINATION_SIZE_MB" FROM V$RECOVERY_FILE_DEST;

NAME                                                                   FILE_DESTINATION_SIZE_MB
---------------------------------------------------------------------- ------------------------
/u01/app/oracle/fast_recovery_area                                                         8900
```

➢ How to disable archive log?

STEP 1: Check the archive log status

```
SQL> archive log list
Database log mode              Archive Mode
Automatic archival             Enabled
Archive destination            USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence     71
Next log sequence to archive   73
Current log sequence           73
```

STEP 2: Change the disable archive log status

```
SQL> shu immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL>
SQL> startup nomount
ORACLE instance started.

Total System Global Area  520091504 bytes
Fixed Size                  8898416 bytes
Variable Size             243269632 bytes
Database Buffers          264241152 bytes
Redo Buffers                3682304 bytes
SQL>
SQL> alter database mount;

Database altered.
```

```
SQL> alter database noarchivelog;
alter database noarchivelog
*
ERROR at line 1:
ORA-38774: cannot disable media recovery - flashback database is enabled


SQL> alter database flashback off;

Database altered.

SQL> alter database noarchivelog;

Database altered.

SQL> alter database open
  2  ;

Database altered.
```

❖ Why Flashback requires ARCHIVELOG mode?

Because Flashback Database uses:

- **Flashback logs**
- **Redo logs**
- **Archive logs**

to rewind the database to a previous point in time.
Without archive logs, Oracle cannot guarantee recovery consistency.

```
SQL> archive log list
Database log mode              No Archive Mode
Automatic archival             Disabled
Archive destination            USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence     72
Current log sequence           74
```

- **How to enable archive log?**

STEP 1: Check the archive log status

```
SQL> archive log list
Database log mode              No Archive Mode
Automatic archival             Disabled
Archive destination            USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence     72
Current log sequence           74
```

STEP 2: Change the enable archive log status

```
SQL> shu immadiate
SP2-0717: illegal SHUTDOWN option
SQL>
SQL>
SQL> shu immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL> startup mount
ORACLE instance started.

Total System Global Area  520091504 bytes
Fixed Size                  8898416 bytes
Variable Size             243269632 bytes
Database Buffers          264241152 bytes
Redo Buffers                3682304 bytes
Database mounted.
```

```
SQL> alter database archivelog;

Database altered.

SQL> alter database flashback on;

Database altered.
```

```
SQL> alter database open;

Database altered.

SQL> archive log list;
Database log mode              Archive Mode
Automatic archival             Enabled
Archive destination            USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence     73
Next log sequence to archive   75
Current log sequence           75
```

- **How to Check size of archive generation per day?**

SELECT
TRUNC(completion_time) AS day,
ROUND(SUM(blocks * block_size)/1024/1024/1024,2) AS size_gb
FROM v$archived_log
GROUP BY TRUNC(completion_time)
ORDER BY day;

```
SQL> SELECT TRUNC(completion_time) AS day,ROUND(SUM(blocks * block_size)/1024/1024/1024,2) AS size_gb FROM v$archived_log GROUP BY TRUNC(compl
etion_time) ORDER BY day;

DAY          SIZE_GB
--------- ----------
04-MAY-26          0
05-MAY-26          0
06-MAY-26        .06
07-MAY-26        .11
08-MAY-26        .03
09-MAY-26        .07
10-MAY-26        .06
11-MAY-26        .01
12-MAY-26        .06
13-MAY-26        .07
14-MAY-26        .06

DAY          SIZE_GB
--------- ----------
15-MAY-26        .03
16-MAY-26        .03
17-MAY-26        .16
18-MAY-26        .04
19-MAY-26        .03
21-MAY-26        .04
22-MAY-26        .03
23-MAY-26        .05
26-MAY-26        .01
27-MAY-26        .11
28-MAY-26        .03

DAY          SIZE_GB
--------- ----------
29-MAY-26        .06
```

➢ **How to Check size of archive generation per hour?**

SELECT
TO_CHAR(completion_time,'DD-MON-YYYY HH24') AS hour,
ROUND(SUM(blocks * block_size)/1024/1024,2) AS size_mb
FROM v$archived_log
GROUP BY TO_CHAR(completion_time,'DD-MON-YYYY HH24')
ORDER BY hour;

```
SQL> SELECT TO_CHAR(completion_time,'DD-MON-YYYY HH24') AS hour,
       ROUND(SUM(blocks * block_size)/1024/1024,2) AS size_mb
FROM v$archived_log
GROUP BY TO_CHAR(completion_time,'DD-MON-YYYY HH24')
ORDER BY hour;  2    3    4    5

HOUR                        SIZE_MB
----------------------- ----------
04-MAY-2026 17                 .01
05-MAY-2026 20                   0
06-MAY-2026 18                  61
07-MAY-2026 09               86.22
07-MAY-2026 12               18.74
07-MAY-2026 14                8.02
08-MAY-2026 11                8.77
08-MAY-2026 17               23.26
08-MAY-2026 18                1.78
09-MAY-2026 10                9.47
09-MAY-2026 15               60.92
```

➢ **How to Check Today's Archive Generation**

SELECT
ROUND(SUM(blocks * block_size)/1024/1024/1024,2) AS today_archive_gb
FROM v$archived_log
WHERE TRUNC(completion_time)=TRUNC(SYSDATE);

```
SQL> SELECT ROUND(SUM(blocks * block_size)/1024/1024/1024,2) AS today_archive_gb
FROM v$archived_log
WHERE TRUNC(completion_time)=TRUNC(SYSDATE);  2    3

TODAY_ARCHIVE_GB
----------------
             .06
```

➢ **How to count archive log per day**

SELECT
TRUNC(completion_time) day,
COUNT(*) total_archives
FROM v$archived_log
GROUP BY TRUNC(completion_time)
ORDER BY day;

```
SQL> SELECT TRUNC(completion_time) day,
       COUNT(*) total_archives
FROM v$archived_log
GROUP BY TRUNC(completion_time)
ORDER BY day;  2    3    4    5

DAY       TOTAL_ARCHIVES
--------- --------------
04-MAY-26              1
05-MAY-26              1
06-MAY-26              1
07-MAY-26              5
```

- **How to check archive log size in OS level?**

```
[oracle@localhost ~]$ du -sh /u01/app/oracle/fast_recovery_area
4.5G    /u01/app/oracle/fast_recovery_area
```

- **How to check FRA usage?**

SELECT name,
ROUND(space_limit/1024/1024/1024,2) AS total_size_gb,
ROUND(space_used/1024/1024/1024,2) AS used_size_gb,
ROUND(space_reclaimable/1024/1024/1024,2) AS reclaimable_gb,
ROUND((space_limit-space_used+space_reclaimable)/1024/1024/1024,2) AS free_space_gb
FROM v$recovery_file_dest;

```
SQL> SELECT name,
  2         ROUND(space_limit/1024/1024/1024,2) AS total_size_gb,
  3         ROUND(space_used/1024/1024/1024,2) AS used_size_gb,
       ROUND(space_reclaimable/1024/1024/1024,2) AS reclaimable_gb,
       ROUND((space_limit-space_used+space_reclaimable)/1024/1024/1024,2) AS free_space_gb
FROM v$recovery_file_dest;  4    5    6

NAME                                                   TOTAL_SIZE_GB USED_SIZE_GB RECLAIMABLE_GB FREE_SPACE_GB
------------------------------------------------------ ------------- ------------ -------------- -------------
/u01/app/oracle/fast_recovery_area                              8.06         3.87           1.03          5.22
```